Sentiment Analysis for E-commerce Reviews

using Machine Learning Techniques.

Case Study Report

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Abstract

This report presents a consolidated NLP study in formal academic structure. The project addresses three tasks: binary sentiment classification with uncertainty control, multi-label issue extraction, and transformer-based extension. The source corpus contains 152,410 reviews; the issue annotation set contains 19,000 rows. In classic sentiment, variant V6 (word 1–2 + char 3–5 + negation, class weight w10) reaches test `precision_0=0.677`, `recall_0=0.932`, and `F2_0=0.867`, with selective coverage 0.885 under threshold band 0.40/0.60. The transformer sentiment extension reaches `accuracy=0.966`, `precision_0=0.799`, `recall_0=0.853`, and `F2_0=0.842`, with coverage 0.910. For issue extraction on the 3,800-row split, OVR Logistic reaches `micro-F1=0.780` and `macro-F1=0.387`, OVR LinearSVM reaches `micro-F1=0.771` and `macro-F1=0.405`, while dummy labelset baselines show high micro-F1 (`0.809`) but very low macro-F1 (`0.101`) due to majority-label dominance. The no-new-data upgrade package adds syllabus baselines, n-gram language modeling, evaluation-rigor artifacts, and transformer issue-routing experiments. All quantitative statements are sourced from saved experiment artifacts under `results/`.

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1 Introduction

User reviews are short, noisy, and often mix praise and complaints in the same sentence. This project treats review understanding as a complete NLP pipeline problem rather than a single classifier benchmark [1, 2]. The system is designed for practical support triage where missing negative feedback has higher operational risk than over-flagging.

1.1 Project Goals

The first goal is to build a sentiment analysis pipeline that can distinguish negative and positive reviews while handling uncertain cases more carefully than a standard hard classifier. The second goal is to move beyond coarse sentiment and extract concrete issue categories that can support downstream service action. The third goal is to compare classic sparse NLP with transformer-based contextual modeling under the same review domain.

1.2 Target Users and Customers

The primary target users are customer-support teams, operations teams, and product analysts working with large volumes of e-commerce feedback. A secondary target group is instructors and students in an NLP course setting, where the project can be used as a case study of how text classification, uncertainty handling, multi-label learning, and transformer extensions connect within one applied problem.

From an application perspective, the problem carries clear operational pain points. First, manual reading at scale is expensive and slow, so urgent negative cases can be buried among large volumes of positive comments. Second, a coarse positive/negative decision is not enough for action: support teams still need explicit issue types such as delivery, redemption, refund, or fraud signals. Third, short and ambiguous reviews create decision risk, where forcing hard labels can increase both false alarms and missed complaints.

These constraints motivate a design that goes beyond single-model accuracy. The project emphasizes recall-sensitive evaluation for risky negative cases, explicit issue extraction for downstream handling, and uncertainty-aware outputs for low-information texts. This framing links NLP model choices directly to practical service quality and response time.

The scope is constrained to no-new-data upgrades: modeling, evaluation, calibration, and analysis improvements are performed on existing data only.

2 NLP Problem Definition

2.1 Task A: Sentiment Understanding

Task A is binary sentiment classification from review text. Ratings 1–2 are mapped to class 0 (negative), ratings 4–5 to class 1 (positive), and rating 3 is reserved as uncertain analysis data. Due to severe imbalance (about 9.5:1 positive to negative), model selection follows a negative-first ordering: maximize `recall_0`, then `F2_0`, then `precision_0`.

2.2 Task B: Issue Extraction

Task B maps review text to a multi-label issue vector with nine labels (delivery, redemption, product quality, customer service, refund, usability, value/price, fraud/scam, other). This converts unstructured text into actionable support categories [4, 5].

2.3 Task C: Deep Context Extension

Task C extends the classic pipeline with transformer fine-tuning for deeper contextual modeling, hard-case comparison, and threshold-behavior analysis [6, 7].

3 Dataset and Language Characteristics

3.1 Data Source and Scale

The source file contains 152,410 reviews. Rating distribution is shown in Table 1.

Table 1: *Rating distribution in source data*

Rating	Count
1	12,326
2	1,873
3	3,271
4	6,692
5	128,248

Table 1 confirms an extreme long-tail sentiment distribution in raw e-commerce reviews. Rating 5 alone accounts for roughly 84.1% of all rows, while rating 1 and 2 together account for about 9.3%. This imbalance explains why accuracy-only evaluation is insufficient and why the project prioritizes negative-class recall and F2. It also

justifies the decision to keep rating 3 as an uncertainty set instead of forcing it into hard supervision.

Strong-label sentiment rows (ratings 1,2,4,5) total 149,139, with 14,199 negatives and 134,940 positives.

3.2 Observed Language Noise

The corpus includes abbreviations (for example *gr8*, *thx*, *idk*), short fragments, typos, and mixed-polarity constructions such as *good but late delivery*. Negation and contrast markers are frequent and motivate dedicated handling in preprocessing and feature design.

3.3 Issue Label Data

The merged issue dataset contains 19,000 annotated rows. Label frequency is shown in Table 2.

Table 2: *Issue label frequency in manual annotation set*

Label	Count	Share
delivery_shipping	972	0.0512
redemption_activation	284	0.0149
product_quality	1,251	0.0658
customer_service	182	0.0096
refund_return	93	0.0049
usability	873	0.0459
value_price	558	0.0294
fraud_scam	144	0.0076
other	15,960	0.8400

The issue distribution in Table 2 is even more skewed than sentiment labels: `other` dominates at 84.0%, and several actionable labels remain below 1.0% prevalence. This creates a high risk that naive models appear strong on aggregate metrics while missing operationally important minority issues. Therefore, macro-F1, per-label F1, and per-label threshold tuning are necessary to avoid majority-label bias in deployment decisions.

4 Text Processing Design

4.1 Normalization

The default cleaner applies lowercasing, URL removal, punctuation spacing, and whitespace normalization. The goal is to reduce sparsity while preserving sentiment-bearing tokens.

4.2 Negation Handling

The classic sentiment and issue pipelines support negation propagation with a local window (current issue config: window size 3). This is critical for polarity flips in short reviews.

4.3 Contrast Handling

Contrast cues such as *but*, *however*, and clause-level polarity shifts are handled indirectly through n-gram features, including phrase patterns that encode polarity transitions.

4.4 Character-level Modeling

Character n-grams (3–5) are combined with word n-grams in selected variants to improve robustness to misspellings, abbreviations, and non-standard surface forms.

5 Feature Engineering

5.1 Classic Sparse Representation

The main sentiment representation is TF-IDF over word n-grams (1–2) and character n-grams (3–5), followed by Chi-square feature selection. The selected best variant (V6) uses `k=10000`.

For issue extraction, TF-IDF with optional character support and Chi-square filtering (selected `k=2000`) is used before one-vs-rest classification.

5.2 Lexicon Feature Block

The current default best models are data-driven sparse models. Lexicon-oriented analyses are retained as optional diagnostic components in the extension package rather than as mandatory blocks in the final deployment path.

6 Modeling Approach

6.1 Sentiment Classifiers

The classic sentiment backbone is Logistic Regression. Class-weight modes are searched across `none`, `balanced`, `w2`, `w5`, and `w10`. Variant search over preprocessing and feature toggles (V0–V7) determines the final configuration.

6.2 Issue Extraction Model

Issue extraction uses calibrated one-vs-rest Logistic Regression with per-label threshold tuning. One-vs-rest LinearSVM and a probability-level blend (`ovr_blend_lr_svm`) are retained as strong classical comparators. Two dummy baselines (`dummy_labelset_majority`, `dummy_label_prior`) are also reported as sanity checks. Current train configuration includes per-label class-weight search and sigmoid calibration (CV=3).

6.3 Transformer Extension

The transformer extension uses HuggingFace sequence-classification fine-tuning (default base family: DistilBERT in this setup). It is evaluated with the same threshold-band policy to enable direct comparison with classic selective behavior.

7 Experimental Setup

7.1 Split and Protocol

Sentiment training uses strong-label data (ratings 1,2,4,5) with held-out validation and test sets; rating 3 is kept for uncertainty analysis. Issue extraction uses a fixed random split of 13,300/1,900/3,800 (train/val/test). The issue-transformer extension reports on 3,000/1,000/2,000.

7.2 Main Metrics

For sentiment, main metrics are `accuracy`, `F1`, `precision_0`, `recall_0`, `F1_0`, and `F2_0`. For issue extraction, metrics are micro-F1, macro-F1, subset accuracy, hamming loss, and per-label F1.

7.3 Uncertainty Rule

Selective classification uses a threshold band $[t_{low}, t_{high}]$. Samples with probability inside the band are marked uncertain; remaining samples are covered decisions. Reported

selective metrics include coverage, selective precision_0, selective recall_0, selective F2_0, and covered false-negative rate [9].

8 No-New-Data Syllabus Upgrade Package

The upgrade package extends the project without collecting new data. Major components in the current experimental package are:

1. Issue Level-3 enhancements: per-label class-weight search, probability calibration, threshold stability diagnostics.
2. Issue Level-4 transformer baseline and hybrid routing with capped route rate.
3. Syllabus benchmark suite: Naive Bayes, Perceptron, LinearSVM, SGD log-loss, semantic SVD baselines, FFNN MLP on SVD features.
4. N-gram language model module with add-k smoothing and perplexity reporting.
5. Evaluation-rigor package: bootstrap confidence intervals, pairwise significance tests, and error taxonomy.
6. Course-fit matrix generated directly from saved artifacts.

9 Results for Classic NLP Sentiment Model

9.1 Baseline Selection by K

Table 3 shows best validation setting per Chi-square dimension under negative-first ordering.

Table 3: *Best sentiment baseline by Chi-square K (validation)*

K	Class weight	Precision_0	Recall_0	F2_0	Accuracy
1,000	w10	0.555	0.910	0.806	0.922
2,000	w10	0.584	0.920	0.825	0.930
5,000	balanced	0.606	0.935	0.843	0.936
10,000	balanced	0.621	0.936	0.850	0.940

Table 3 shows a monotonic quality gain as feature dimension increases from 1,000 to 10,000. Precision_0 improves from 0.555 to 0.621 and F2_0 rises from 0.806 to

0.850, while recall_0 stays high across all settings. This pattern indicates that larger sparse vocabularies mainly improve false-positive control on the negative class without sacrificing triage sensitivity.

9.2 Variant Comparison

Table 4 reports test performance for engineered variants. V6 is the selected main model.

Table 4: *Classic sentiment variant comparison on test*

Variant	K	Class weight	Precision_0	Recall_0	F2_0
V0	10,000	balanced	0.638	0.937	0.856
V1	10,000	balanced	0.643	0.932	0.855
V2	5,000	w10	0.647	0.926	0.852
V3	10,000	balanced	0.626	0.923	0.843
V4	10,000	balanced	0.635	0.921	0.845
V5	10,000	balanced	0.653	0.933	0.859
V6	10,000	w10	0.677	0.932	0.867
V7	5,000	w10	0.657	0.927	0.857

Compared with dummy sentiment baselines (`dummy_most_frequent: F2_0=0.000`, `dummy_stratified: F2_0=0.092`), the selected V6 model (`F2_0=0.867`) shows a large practical margin on the negative class objective. Within the non-dummy family, V6 also gives the best balance: it achieves the highest precision_0 (0.677) while retaining recall_0 above 0.93, outperforming nearby variants V5 and V7 on F2_0. This suggests that the joint use of word+character TF-IDF plus negation modeling contributes complementary signals rather than redundant complexity.

9.3 Threshold and Uncertainty Behavior

For V6, the recommended threshold band is 0.40/0.60. Table 5 shows sensitivity to threshold movement.

Table 5: *Classic sentiment threshold sweep on test*

Thresholds	Coverage	Sel. precision_0	Sel. recall_0	Sel. F2_0
0.40 / 0.60	0.885	0.730	0.956	0.900
0.45 / 0.60	0.891	0.708	0.956	0.894
0.50 / 0.65	0.886	0.681	0.963	0.889
0.55 / 0.70	0.881	0.646	0.970	0.882

Table 5 reveals a controlled precision-recall trade-off as thresholds move upward. Increasing the positive-decision band improves selective recall_0 (0.956 to 0.970) but reduces selective precision_0 (0.730 to 0.646), while coverage remains stable around 0.88–0.89. This stability is valuable for operations because policy adjustments can tune risk profile without large throughput disruption.

10 Results for Transformer NLP Extension

10.1 Transformer Metrics

Table 6 summarizes transformer sentiment metrics.

Table 6: *Transformer sentiment performance*

Split	Accuracy	F1	Precision_0	Recall_0	F2_0	Coverage
Validation	0.966	0.981	0.793	0.870	0.854	0.906
Test	0.966	0.981	0.799	0.853	0.842	0.910

Table 6 indicates strong generalization consistency: validation and test metrics are closely aligned, with no sign of major overfitting in one-epoch fine-tuning. Compared to the classic model, transformer improves negative precision materially (0.799 vs 0.677) but reduces negative recall (0.853 vs 0.932). This means transformer is better at avoiding false alarms, while classic V6 remains better at catching risky negative cases under the project's triage-first objective.

10.2 Transformer Threshold Sweep

Table 7: *Transformer threshold sweep on test*

Thresholds	Coverage	Sel. precision_0	Sel. recall_0	Sel. F2_0
0.40 / 0.60	0.910	0.831	0.846	0.843
0.45 / 0.60	0.911	0.827	0.847	0.843
0.50 / 0.65	0.912	0.821	0.850	0.844
0.55 / 0.70	0.911	0.818	0.855	0.847

Table 7 shows the transformer selective curve is relatively flat, with coverage near 0.91 across all tested bands and only small F2_0 movement (0.843 to 0.847). This suggests the model is calibration-stable in the evaluated range and can be tuned conservatively without destabilizing decision volume.

10.3 Hard-case Comparison

Saved hard-case probes show complementary behavior. For *not bad* and *good but late delivery*, the classic model predicts negative while the transformer predicts positive. For very short tokens (*gr8*, *thx*, *idk*), both systems route to uncertain due to short-text policy.

11 Results for NLP Issue Extraction

11.1 Overall Multi-label Results

Table 8 reports the current main pipeline results (test size 3,800).

Table 8: *Issue extraction overall metrics*

Model	Micro-F1	Macro-F1	Subset accuracy	Hamming loss
OVR Logistic (test)	0.780	0.387	0.634	0.0585
OVR LinearSVM (test)	0.771	0.405	0.617	0.0623
OVR Blend LR+SVM (test)	0.768	0.391	0.626	0.0620
Dummy labelset majority (test)	0.809	0.101	0.811	0.0438
Dummy label prior (test)	0.809	0.101	0.811	0.0438

Dummy baselines rank high on micro-F1 and subset accuracy because they almost always predict the dominant `other` label; macro-F1 exposes that they fail on minority issue labels. From Table 8, OVR Logistic is preferred for balanced production utility: it leads micro-F1 and subset accuracy among non-dummy models, while LinearSVM offers slightly better macro-F1. The blend does not beat both parents, implying that simple averaging is not sufficient under the current label imbalance and would require stronger routing or calibration logic.

11.2 Per-label Improvement

Table 9 compares per-label F1 between SVM and Logistic on test.

Table 9: *Per-label F1 change (SVM minus Logistic)*

Label	Logistic F1	SVM F1	Δ
delivery_shipping	0.442	0.446	+0.004
redemption_activation	0.298	0.287	-0.010
product_quality	0.307	0.282	-0.025
customer_service	0.225	0.168	-0.057
refund_return	0.435	0.387	-0.048
usability	0.459	0.462	+0.003
value_price	0.258	0.313	+0.056
fraud_scam	0.150	0.391	+0.241
other	0.911	0.911	+0.001

Table 9 highlights where SVM helps versus hurts. The largest gain is on `fraud_scam` (+0.241), followed by `value_price` (+0.056), indicating better separation on specific sparse semantic cues. However, Logistic remains better on `customer_service`, `refund_return`, and `product_quality`, which are critical support labels; this is why the main pipeline keeps Logistic as default and uses SVM as a comparator rather than replacement.

11.3 Threshold Strategy in Multi-label Setting

The tuned per-label thresholds for the main `ovr_logreg` pipeline are shown in Table 10. Lower thresholds are assigned to rare labels to improve recall, while `fraud_scam` and `other` are kept more conservative.

Table 10: *Final per-label thresholds in issue pipeline*

Label	Threshold
delivery_shipping	0.25
redemption_activation	0.10
product_quality	0.10
customer_service	0.10
refund_return	0.15
usability	0.10
value_price	0.15
fraud_scam	0.50
other	0.30

Threshold values in Table 10 encode an explicit policy: rare but important labels are granted low activation cutoffs (0.10–0.15) to reduce missed incidents, while high-risk false-positive channels are constrained with higher cutoffs (`fraud_scam`=0.50, `other`=0.30). This per-label asymmetry is more realistic than a global threshold and better aligned with downstream ticketing workflows.

12 Results for Syllabus Upgrade Package

12.1 Classic Baselines for Core Course Algorithms

Table 11 summarizes no-new-data sentiment baselines.

Table 11: *Syllabus benchmark baselines on test*

Model	Recall_0	Precision_0	F2_0
logreg_l2	0.913	0.703	0.862
multinomial_nb	0.913	0.633	0.839
complement_nb	0.952	0.510	0.811
perceptron	0.828	0.684	0.795
linear_svm	0.847	0.789	0.835
sgd_log_loss	0.921	0.616	0.838
svd_semantic_logreg	0.920	0.501	0.788
ffnn_mlp_svd	0.730	0.799	0.742

Table 11 provides a clear algorithmic trade-off map for defense: Complement NB maximizes recall_0 (0.952) but with weak precision_0 (0.510), while LinearSVM gives high precision_0 (0.789) at lower recall_0. The best negative-class balance under F2 is achieved by logistic regression (`F2_0=0.862`), which supports its role as the primary classic baseline in this project.

12.2 Selective Behavior in the Added Baselines

Table 12: *Selective metrics for probabilistic syllabus baselines (test)*

Model	Coverage	Sel. precision_0	Sel. recall_0	Sel. F2_0
logreg_l2_selective	0.887	0.763	0.946	0.903
multinomial_nb_selective	0.903	0.654	0.928	0.857
complement_nb_selective	0.900	0.545	0.963	0.835
sgd_log_loss_selective	0.863	0.696	0.959	0.892

Selective results in Table 12 show that `logreg_l2_selective` remains strongest on Sel. F2_0 (0.903), combining high Sel. recall_0 (0.946) with solid precision_0 (0.763). Complement NB still reaches the highest Sel. recall_0 (0.963) but precision collapses, confirming it is suitable only when maximum capture is prioritized over review workload.

12.3 N-Gram Language Model Results

Table 13: *N-gram LM perplexity (add-k smoothing)*

Model	Validation perplexity	Test perplexity
Unigram add-k	376.51	378.02
Bigram add-k	358.59	363.08

Table 13 confirms expected language-model behavior: bigram consistently outperforms unigram on both validation and test perplexity. On test data, perplexity drops from 378.02 to 363.08 (about 3.95% relative reduction), showing that even short review texts benefit from one-step context modeling. This provides concrete evidence for the course N-gram component beyond classification metrics.

12.4 Updated Course-Fit Matrix

Table 14 reports the topic-coverage matrix generated from current artifacts. The mean coverage is **94.3%**.

Table 14: *Course-fit matrix from current artifact set*

Topic	Coverage (%)
Probability	100
Regular Expressions	50
Evaluation Measures	100
N-Gram Language Models	100
Naive Bayes	100
Perceptron	100
Logistic Regression	100
Feed-forward Neural Networks	100
Pytorch Basic	100
Vector Semantics and Embeddings	80
Recurrent Neural Networks	100
Transformers and LLMs	100
Masked Language Models	100
Applications of LLMs	90

The matrix in Table 14 indicates strong breadth across core topics, with most items at 100%. Remaining gaps are concentrated in *Regular Expressions* (50%) and partly in *Vector Semantics and Embeddings* (80%), so these are the highest-impact targets if the team wants to further improve academic completeness without changing dataset scope.

13 Classic Versus Transformer Discussion

On sentiment classification, transformer improves overall accuracy and negative precision (0.799 vs 0.677), while classic V6 remains stronger on negative recall (0.932 vs 0.853) and F2_0 (0.867 vs 0.842). Under the same threshold band (0.40/0.60), transformer yields higher selective precision but lower selective recall than classic.

For issue extraction, Table 15 reports the dedicated fair-split benchmark. The classic issue model remains strongest; hybrid routing improves over pure transformer but still trails the classic setting.

Table 15: *Issue extraction fair-split comparison*

Model	Micro-F1	Macro-F1	Subset accuracy	Hamming loss
classic_issue_model	0.792	0.528	0.656	0.0549
hybrid_route	0.730	0.338	0.615	0.0740
transformer_multilabel	0.646	0.188	0.551	0.1013

Relative to pure transformer on this fair split, the classic issue model gains +0.146 micro-F1 and +0.340 macro-F1. Table 15 also shows that hybrid routing recovers part of the transformer gap but remains clearly below the classic model, especially on macro-F1 where minority-label handling is critical. This suggests transformer issue modeling is not yet data-efficient enough under current annotation volume and should be treated as an auxiliary component until label support increases or architecture is adapted.

14 Error Analysis and Failure Modes

The evaluation-rigor package records 1,454 sentiment misclassified rows (194 false negatives for class 0, 1,260 false positives for class 0). Dominant false-positive categories are negation patterns (49.8%), other mixed contexts (32.4%), and contrast patterns (10.4%). Dominant false-negative categories are other contexts (52.1%), short text (23.2%), and negation patterns (14.4%).

Pairwise significance tests against baseline comparators show positive deltas for V6 (for example recall_0 +0.063 vs multinomial NB, +0.085 vs LinearSVM), but p-values remain above 0.05 in this run, so gains are treated as directional rather than statistically confirmed.

15 Practical Value for NLP Course

The project now provides a full, reproducible course artifact chain: cleaned datasets, stepwise experimental outputs, saved models, threshold analyses, uncertainty policy, transformer extension, and evaluation-rigor outputs. This supports both engineering deployment discussion and academic defense requirements.

The upgraded syllabus package strengthens mapping to core course topics while preserving the same data budget, making the project suitable for methodology-focused assessment. Based on measurable experimental artifacts, the latest rubric assessment estimates 18.04/20 (90.2%) and syllabus-topic coverage 94.3%.

16 Limitations and Future Work

Main limitations are class imbalance, high dominance of the `other` issue label, and weak transformer multi-label performance under current small-split fine-tuning. Error taxonomy also shows persistent sensitivity to negation and mixed-contrast sentences.

Future work should prioritize calibration-aware ensembling, better contextual augmentation without new labels, and lightweight sequence models for issue extraction that preserve interpretability.

17 Conclusion

This report summarizes model structure and quantitative findings in one consistent analysis. The main production model for sentiment is still the classic V6 variant because it optimizes negative recall/F2 under project policy. Transformer sentiment provides stronger precision and overall accuracy, and remains valuable for complementary analysis. For issue extraction, calibrated classic OVR models remain the strongest option in the current experiments. The no-new-data upgrade package substantially improves academic completeness, reproducibility, and course-topic coverage.

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https://arxiv.org/abs/1705.08500

Appendix A: Sentiment Figures

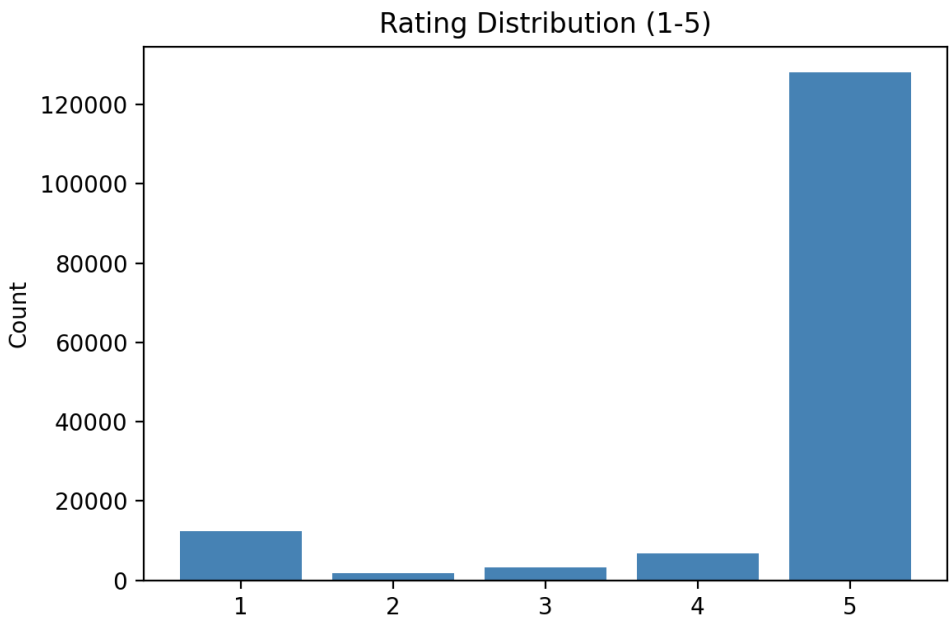

Figure 1: *Rating distribution in source data*

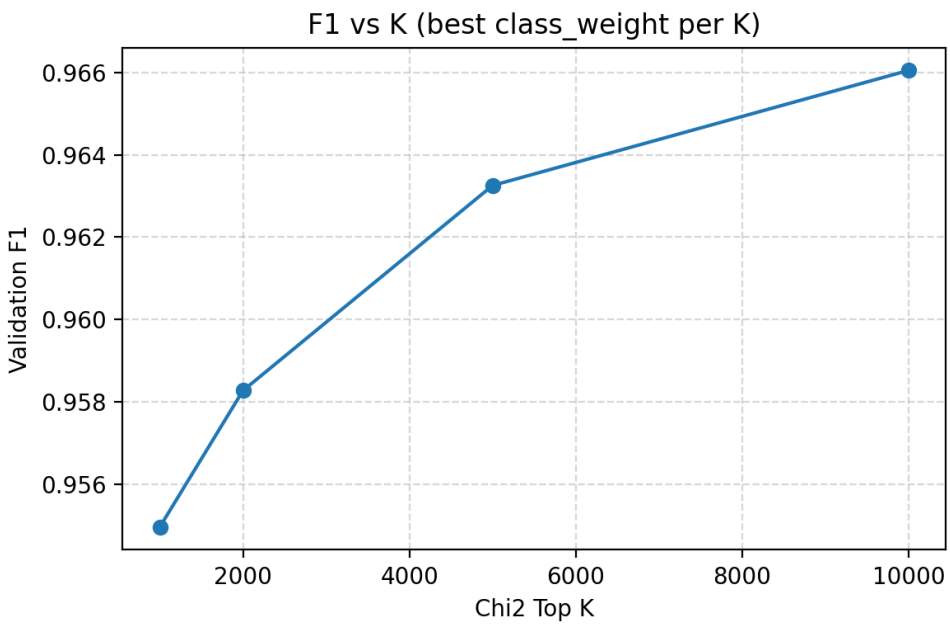

Figure 2: *Validation F1 by Chi-square top K*

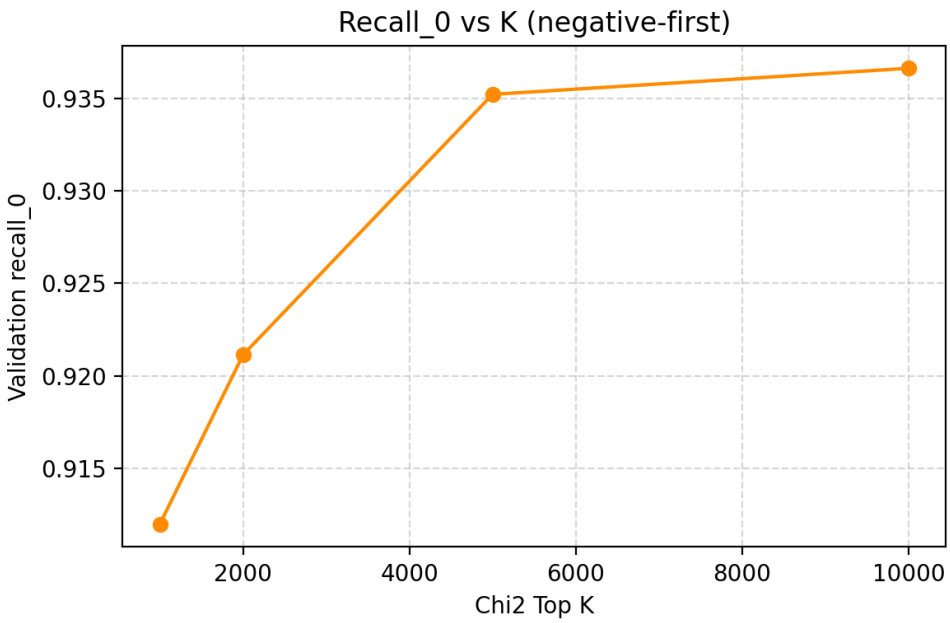

Figure 3: *Validation recall for negative class by top K*

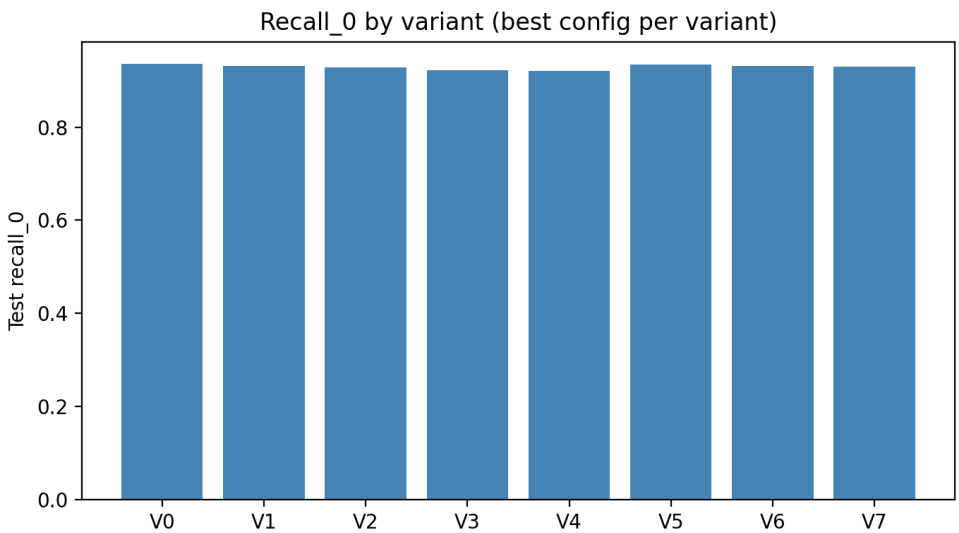

Figure 4: *Test recall for negative class by classic variant*

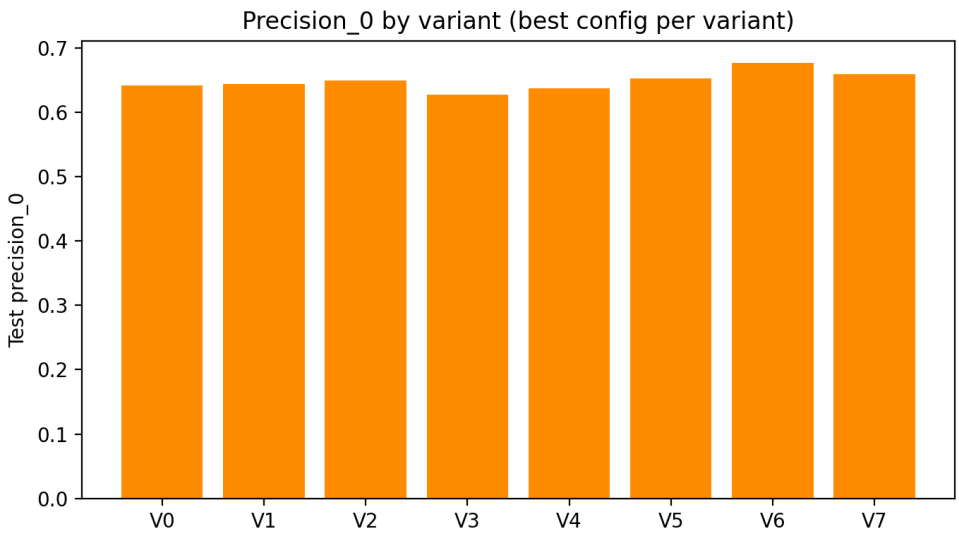

Figure 5: *Test precision for negative class by classic variant*

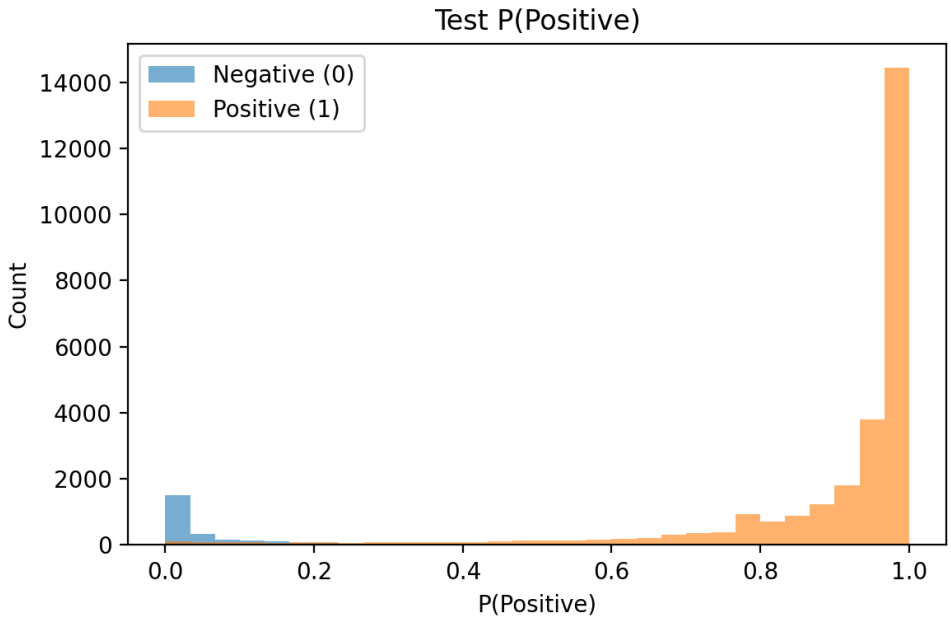

Figure 6: *Classic probability histogram on strong test set*

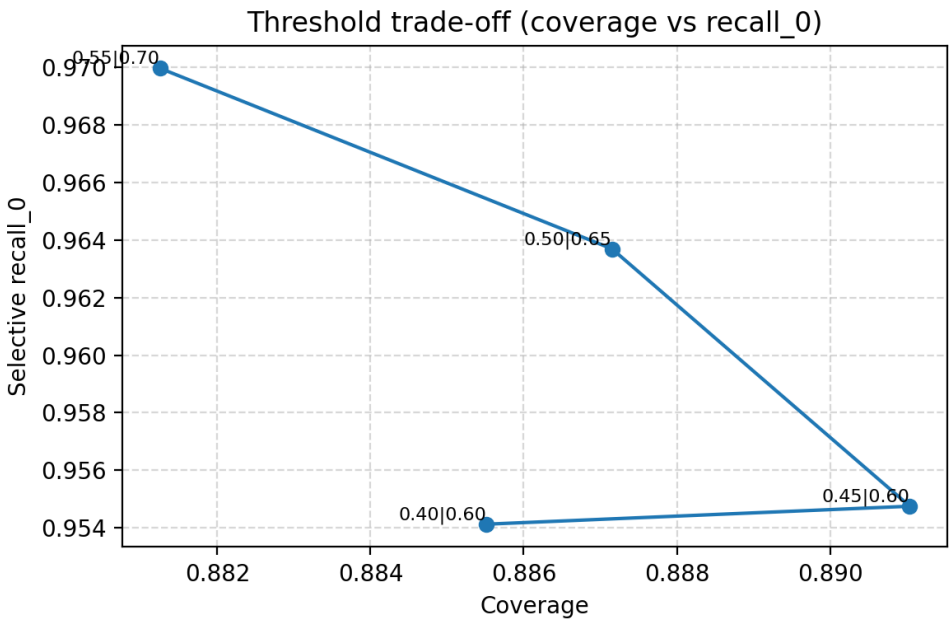

Figure 7: *Classic threshold trade-off for coverage and recall*

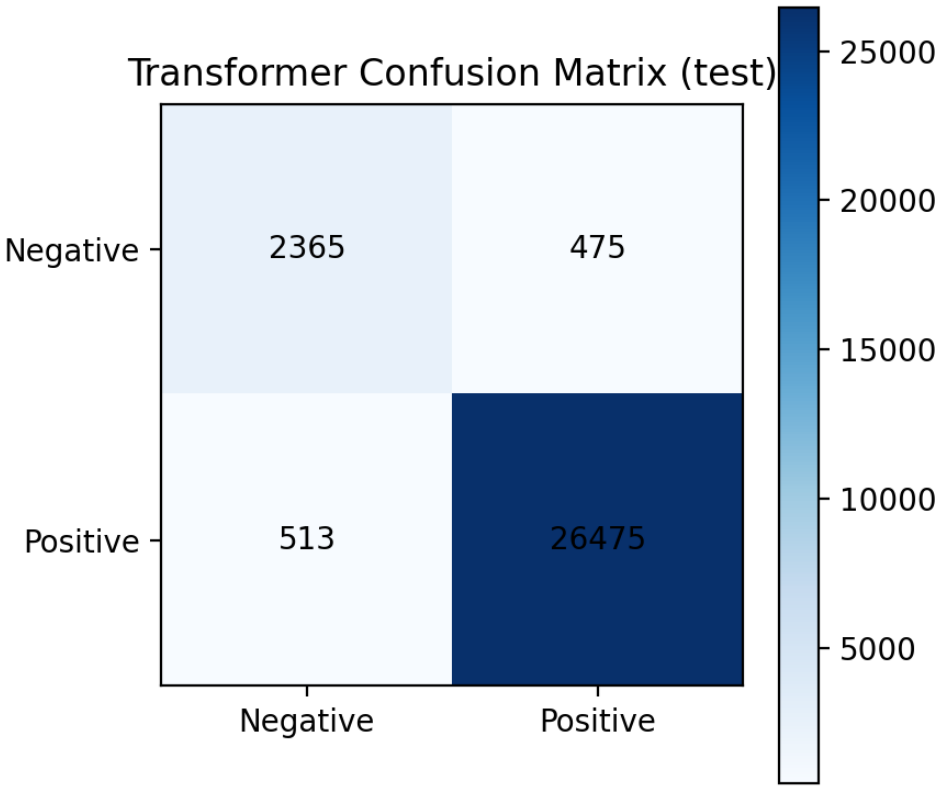

Figure 8: *Transformer confusion matrix on test set*

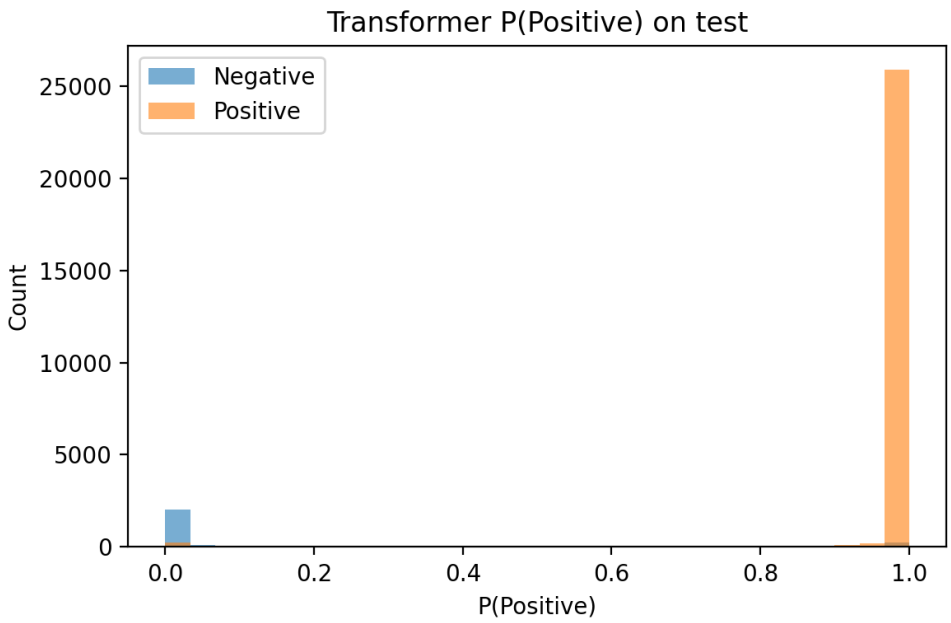

Figure 9: *Transformer probability histogram on test set*

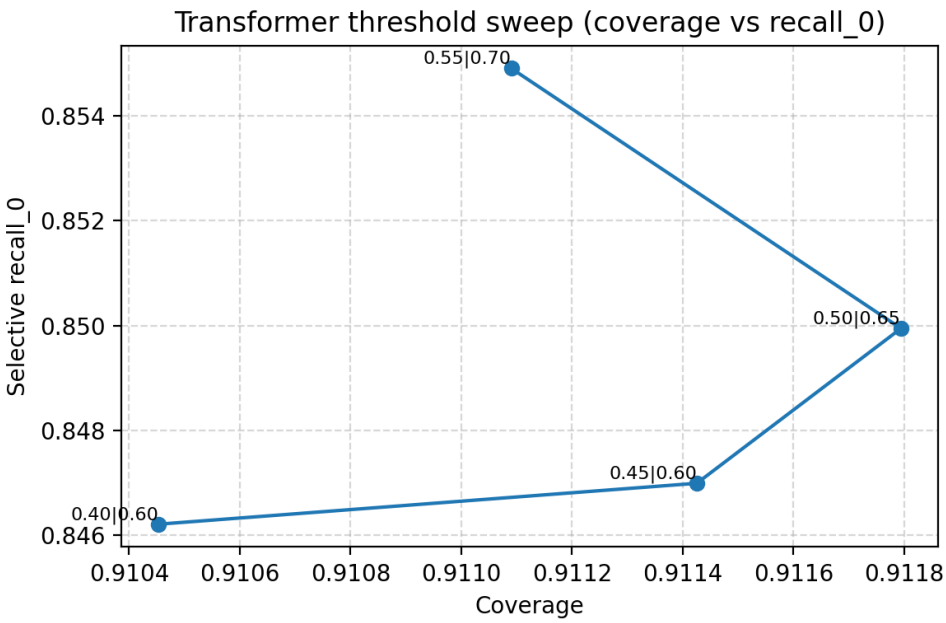

Figure 10: *Transformer threshold trade-off for coverage and recall*

Appendix B: Issue Extraction Figures

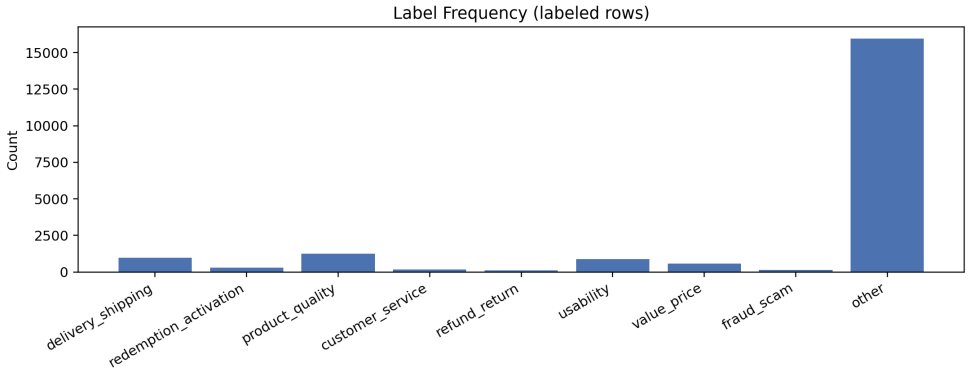

Figure 11: *Issue label frequency in annotation set*

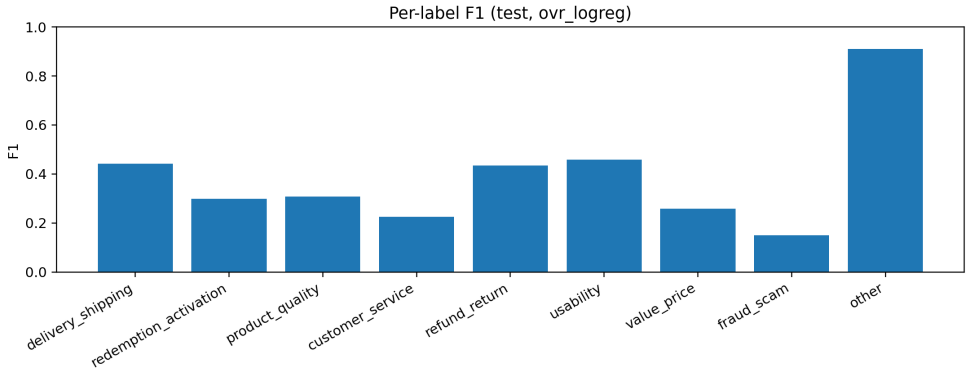

Figure 12: *Per-label F1 for OVR Logistic issue model*

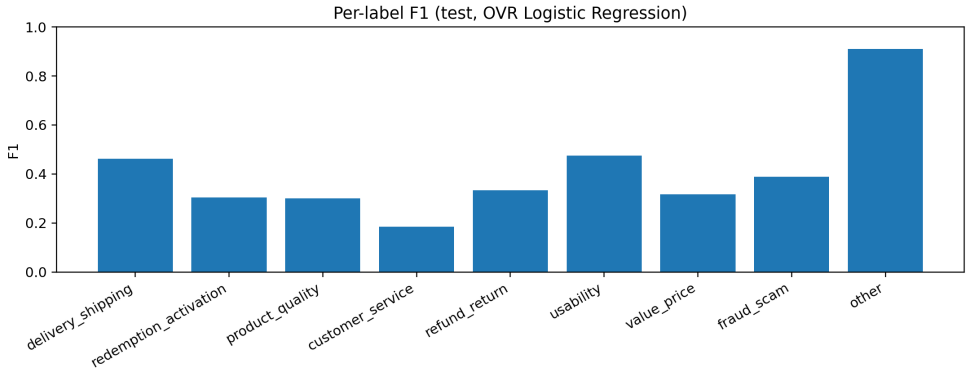

Figure 13: *Per-label F1 for word+char issue variant*

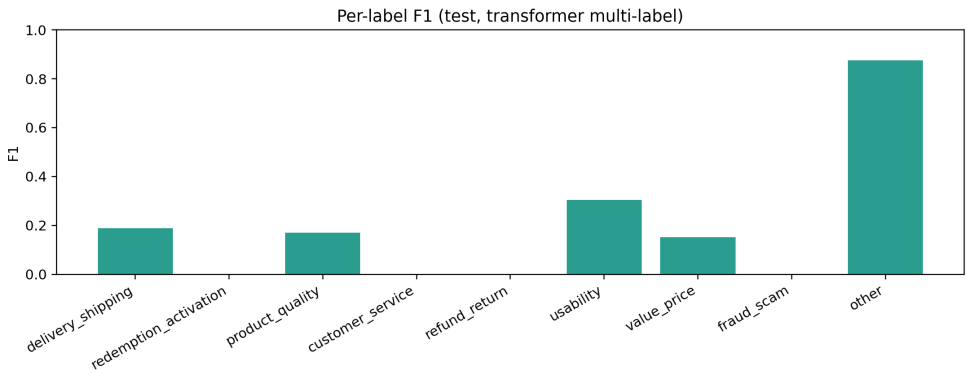

Figure 14: *Per-label F1 for transformer issue baseline*

Appendix C: Syllabus Upgrade Figures

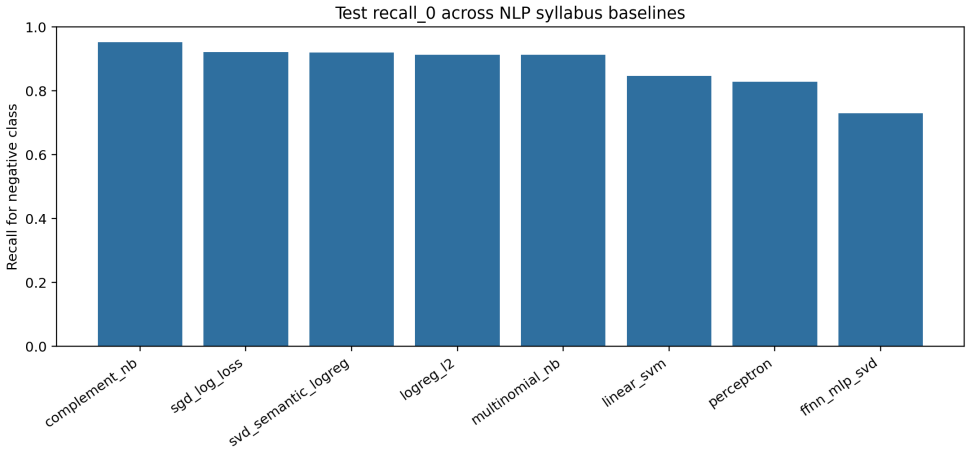

Figure 15: *Test recall_0 across syllabus baselines*

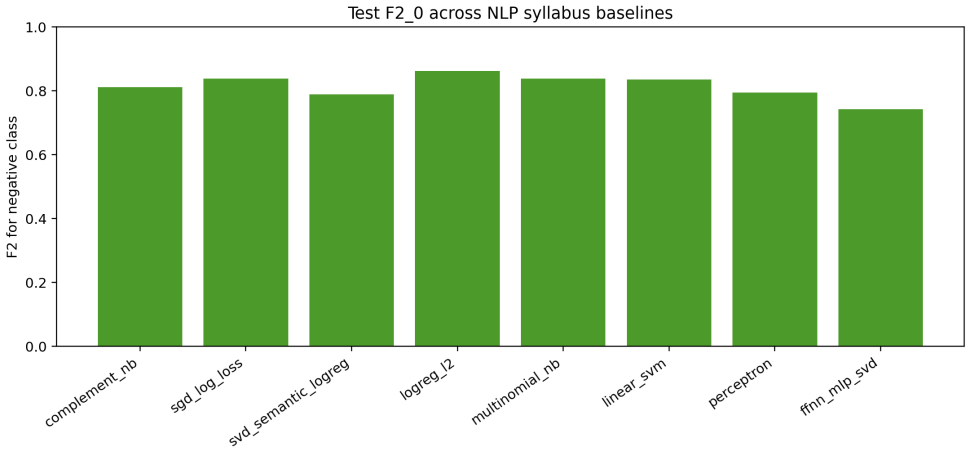

Figure 16: *Test F2_0 across syllabus baselines*